

Theorem 1:

If $a, b, c \in \mathbb{Z}^+$,

the Diophantine equation

$$ax + by = c$$

$\hookrightarrow$ has an integer solution

$x = x_0, y = y_0$, iff

~~if and only if~~ $\gcd(a, b)$ divides c .

$$d = \gcd(a, b).$$

$\Rightarrow$ Assume that $d \mid c$.

$$\Rightarrow c = kd.$$

Bézout's identity: $ax + by = d$.

Multiply both sides by k .

$$\therefore kax + kby = kd = c$$

$$a(\underbrace{kx}_{x_0}) + b(\underbrace{ky}_{y_0}) = c.$$

$\swarrow$ integers.

$\Leftarrow ax + by = c$ indeed has an integer solⁿ

$$d = \gcd(a, b)$$

$$d|a \quad \& \quad d|b.$$

For any integer linear combination of a, b :

$$d|ax + by$$

where x & y are integers.

$$\Rightarrow d|c.$$

□.

Definition:

For $a, b, c \in \mathbb{Z}^+$, c is a common multiple of a, b if c is a multiple of both a & b . Also, c is the least common multiple if c is the lowest among all the common multiples of a, b .

Theorem 2.

Let $a, b, c \in \mathbb{Z}^+$ and $c = \text{lcm}(a, b)$
If d is a cm of a, b then c/d .

Let us assume the statement is not true.

By way of contradiction:

$$d = qc + r, \quad 0 < r < c.$$

$$\because c = \text{lcm}(a, b) \Rightarrow c = ma.$$

$$\text{also } d = na.$$

$$na = qma + r$$

$$(n - qm)a = r > 0.$$

r is a multiple of a .

r is a multiple of b also (similarly).

$\therefore r$ is a 'cm' of a, b . & $r < c$.

$\therefore c$ cannot be the 'lcm'. contradiction $\square$.

Theorem 3: (H.W.)

For $a, b \in \mathbb{Z}^+$, prove that $ab = \text{lcm}(a, b) \cdot \text{gcd}(a, b)$.

Lemma 1.

If $a, b \in \mathbb{Z}^+$ and p is prime, then
 $p \mid ab \Rightarrow p \mid a \text{ or } p \mid b$.

If $p \mid a \rightarrow$ we are done.

If not, $\text{gcd}(p, a) = 1$

$$\Rightarrow ax + py = 1$$

Multiply b on both sides:

$$\underbrace{abx}_{p \mid ab} + \underbrace{pby}_{p \mid p} = \underbrace{b}_{p \mid b}$$

□.

$$15x + 12y = 107$$

E.g. 1.

Show that $\sqrt{2}$ is irrational.

By way of contradiction.

If not then $\sqrt{2} = \frac{a}{b}$ and $\gcd(a, b) = 1$.
↳ co-primes.

$$\Rightarrow \frac{a^2}{b^2} = 2$$

$$\Rightarrow a^2 = 2b^2$$

$$\Rightarrow 2 \mid a^2 \Rightarrow 2 \mid a \quad (\text{Lemma 1}).$$

$$\Rightarrow a = 2c \quad \text{for } c \in \mathbb{Z}^+$$

$$2b^2 = a^2 = (2c)^2 \Rightarrow b^2 = 2c^2$$

$$\Rightarrow 2 \mid b^2$$

$$\Rightarrow 2 \mid b \quad (\text{Lemma 1}).$$

but $\gcd(a, b) = 1$ Contradiction.

Corollary: $\sqrt{p}$ is irrational for any prime p .

Theorem 3

The fundamental theorem of arithmetic.

Every integer $n > 1$ can be written as a product of primes and this factorization is unique.

Strong induction as the proof technique.

Existence:

Base case : 2.

product of one prime.

I. Hypothesis: Suppose for any integer $1 < k < n$ we can write k as a product of primes.

I. Step: If n is prime we are done.
If n is composite we can write
 $n = ab$ s.t. $1 < a, b < n$.

By I.H. $\therefore a = p_1 \cdots p_i$ $b = p'_1 \cdots p'_j$
 $\therefore n = ab = p_1 \cdots p_i p'_1 \cdots p'_j$

Uniqueness.

Base: 2 has only ~~one~~ way of factorisation
 $\Rightarrow$ unique.

I.H.: For every integer $1 < k < n$, k
has a unique prime factorization.

Suppose $n = p_1 p_2 \dots p_r = q_1 q_2 \dots q_s$ (p_i, q_j
are all primes).

Our objective is to show that
 $r = s$ and $\{p_i\} = \{q_j\}$ are
identical.

S1:

$$p_1 | n = q_1 \dots q_s$$

$$\Rightarrow p_1 | q_j$$

But q_j is prime.

$$\therefore p_1 = q_j$$

WLOG, we re order

$$\text{and say } p_1 = q_1$$

Divide $\frac{n}{p_1} = q_2 \cdots q_s = p_2 \cdots p_r$.

By strong induction:

$\frac{n}{p_1}$ is unique.

$$r-1 = s-1$$

$\downarrow$

$$r = s$$

$\frac{n}{p_1}$ is unique & we ~~have~~ have

also got $p_1 = q_1$

$\therefore n$ is unique.

□.

Pigeon hole principle: (PHP).

⇒ If m pigeons occupy n pigeonholes and $m > n$ then at least one hole must be shared by two or more pigeons.

E.g. 1. A tape contains 500,000 "words".
Words can be of four or fewer letters (repetition allowed).
Consecutive words are ~~separated~~ separated by blank.

Only English lower case alphabet constitute the letters.

Can all words be distinct?

$$26^4 + 26^3 + 26^2 + 26 = 475,254.$$

By PHP not all words can be unique. □.